NAVKAR SOLUTIONS

OWNLY - AI RETAIL MANAGER

Complete Feature Catalogue and Week-wise Development Plan

Initial Target Segment: Small and Medium Clothing Retail Stores

Project Completion Deadline: 30 September 2026

1. Product Vision and Positioning

Ownly - AI Retail Manager is a scalable retail-management platform intended to support multiple types of retail businesses. The first release will target small and medium-sized clothing retailers because clothing inventory requires size-wise stock, individual-item identification, fast barcode billing and careful restocking decisions.

Account Onboarding Rule

During account creation, the owner will select the retail-business type. Clothing and Fashion will be enabled in the first release. Other retail types may appear as Coming Soon, while the underlying architecture remains configurable for future expansion.

1.1 Retail Type Selection

- Clothing and Fashion - enabled in the first release.
- Footwear - future template.
- Grocery - future template.
- Electronics - future template.
- Cosmetics and Beauty - future template.
- Jewellery - future template.
- General Retail / Other - future configurable template.

1.2 Clothing Template Enabled After Selection

- Clothing categories and subcategories.
- Size-wise inventory and quantity entry.
- Individual-piece barcode generation.
- Optional product-photo analysis and visual fingerprinting.
- Size, style, product-performance and restocking analytics.
- Clothing-specific dashboard cards and reports.

1.3 Project Objectives

- Digitize account, staff, product, supplier, stock, billing, return and restocking workflows.
- Track inventory at product, size, variant and individual physical-item level.
- Generate one unique barcode for every physical item added or restocked.
- Make billing faster and more accurate through barcode scanning.
- Use actual store transactions to analyse sales speed, size demand, profitability and dead stock.
- Support manual business operations even when AI is not used.
- Provide optional offline AI for product analysis and restocking decisions.
- Avoid a mandatory paid database or paid AI-service dependency.
- Create a scalable foundation that can later support other retail types, multi-store use and mobile/desktop applications.

2. Complete Feature Catalogue

2.1 Account, Store and Onboarding

- Owner account creation using name, phone/email and password.
- Account verification and secure login.
- Retail-business type selection with Clothing and Fashion enabled.
- Store name, address, phone, optional GST details, currency, tax and invoice settings.
- Store logo and branding, optional.
- Automatic Owner role assignment.
- Guided first-time setup checklist.

2.2 Staff, Roles and Permissions

- Owner, Manager, Cashier, Inventory Staff, Accountant and Viewer roles.

- Permission control for cost price, profit, discounts, stock edits, returns, reports, staff management and backups.
- Owner approval for sensitive actions such as invoice cancellation or cost-price changes.
- Login history, session controls and role-based navigation.

2.3 Product and Supplier Setup

- Create product categories and subcategories.
- Create and select suppliers.
- Supplier name mandatory; contact, address and payment terms optional.
- Retail-template-driven product fields so future retail categories can use different attributes.

2.4 Product Entry

- Manual Entry and Optional AI-Assisted Photo Entry.
- Mandatory: category, subcategory, supplier, cost price, selling price and size-wise quantity.
- Size checkboxes with a separate quantity field beside every selected size.
- Automatic total quantity, purchase value, expected sales value and potential gross profit.
- Optional: product name, colour, brand, pattern, style, material, season, gender, tax, description and photo.

2.5 Optional AI Product Photo Capture

- Capture or upload a product photo from phone, tablet or PC.
- Create a numerical visual fingerprint / image embedding.
- Suggest category, subcategory, visible colour, pattern and style.
- Compare with existing products and warn about possible duplicates.
- User must review and approve all suggestions.
- Manual product entry remains fully available.

2.6 Product, Variant and Item Identity

- Product record for the common design.
- Product Variant record for each size or other variation.
- Inventory Item record for each physical piece.
- Variant SKU shared by similar size items, while every physical item has a unique barcode.

2.7 Barcode Generation and Printing

- Generate one unique barcode for every physical quantity.
- If quantity is 20, generate exactly 20 unique barcodes.
- Barcode label shows product name, size, selling price, SKU and unique barcode.
- Thermal printer, A4 labels, selective printing and print-all support.
- Barcode reprint without creating duplicate stock.
- Manual barcode entry fallback.

2.8 Inventory Management

- Statuses: Available, Sold, Returned, Exchanged, Damaged, Defective, Missing and Supplier Return.
- Search by product, SKU, barcode, category, subcategory, size, supplier and status.
- Item-level movement history and complete audit trail.
- Category-, supplier- and size-wise stock views.
- Total cost value, expected selling value and potential gross profit.
- Low-stock, out-of-stock and stock-ageing alerts.
- Slow-moving and dead-stock indicators.

2.9 Barcode Billing and Payments

- Scan using phone/tablet camera, USB scanner(optional), Bluetooth scanner(optional), or manual entry.
- Automatically load product name, size and selling price.
- Authorized discount control.
- Optional customer name and phone.
- Cash, UPI, card and mixed payment.

- Printable and digital invoice.
- Mark individual barcodes Sold and update inventory immediately.

2.10 Customer and Invoice Management

- Optional customer profile and purchase history.
- Invoice number, store information, products, size, price, discount, tax and payment mode.
- Thermal and A4 invoice formats.
- Controlled invoice cancellation and audit logging.
- Future loyalty, membership and personalized recommendations.

2.11 Return and Exchange Management

- Scan the sold barcode and locate the original invoice.
- Capture return/exchange reason and item condition.
- Support refund, credit note and price difference.
- Return eligible item to stock or mark damaged, defective or supplier return.
- Use return and exchange data for product, supplier and size-risk analysis.

2.12 Manual Restocking

- Select an existing product or scan an item barcode.
- Confirm supplier, cost price and selling price.
- Select sizes and enter a restock quantity for each.
- Review total quantity and purchase value.
- Generate a fresh unique barcode for every new physical item.
- Preserve previous product sales and AI history.

2.13 AI-Assisted Restocking

- Capture or upload a garment photo and enter supplier cost, price, sizes and planned quantity.
- Compare with similar historical products.
- Show demand, selling, profitability, local-fit, dead-stock and return-risk scores.
- Estimate sales speed, 30-day sales and stock-clearance period.
- Recommend total purchase quantity and size-wise mix.
- Show confidence and written reasons.
- Owner can accept, modify, ignore or reject the recommendation.

2.14 Offline AI Product Performance

- Time to first sale, sales velocity and sell-through rate.
- Units sold in 7, 15 and 30 days.
- Size-wise demand and price-band performance.
- Profitability, discount dependency and return rate.
- Dead-stock probability and product lifecycle stage.
- Recommended actions: Restock, Hold, Controlled Buy, Trial, Discount, Clearance or Avoid.
- Prediction-versus-actual tracking for continuous improvement.

2.15 Additional Offline AI Features

- Similar-product performance comparison.
- Duplicate product detection.
- Price recommendation and profit simulation.
- Discount recommendation.
- Dead-stock risk detection.
- Size distribution recommendation.
- Supplier performance score.
- Customer segmentation, future phase.
- Billing and inventory anomaly alerts.
- Product-photo quality validation.
- Optional OCR for size, MRP or product labels, advanced phase.

2.16 Dashboard and Reports

- Today, week, month and year sales.
- Gross profit, margin, average bill value, invoice count and items sold.
- Payment-mode distribution and period comparison.
- Available stock, stock cost value, expected sales value and potential profit.
- Low stock, out of stock, damaged, missing, returned and ageing stock.
- Top-selling, slow-moving, highest-profit and high-return products.
- Fastest-selling sizes and best categories.
- AI restock recommendations, selling probability, dead-stock risk and expected sales.
- Supplier, customer and staff insights.
- Quick actions: Add Product, Bill, Scan, Restock, AI Analysis, Inventory, Labels, Export and Backup.

2.17 Reports and Exports

- Sales, profit, inventory, stock valuation, stock ageing and dead-stock reports.
- Category, subcategory, product, size and supplier performance reports.
- Returns, exchanges, staff, payment mode and audit reports.
- AI recommendation and prediction-accuracy report.
- Filter by date, category, supplier, product, size, cashier and status.
- CSV, Excel and printable export.

2.18 Backup, Restore and Data Safety

- Automatic local backup and manual full-backup export.
- Restore from backup with validation.
- Inventory, sales, customer and AI-data export.
- Model version backup and rollback.
- Storage-health indicator and backup reminders.
- Protected deletion and recovery procedures.

2.19 Responsive PWA and Future Applications

- Phone, tablet, laptop and desktop layouts.
- Add to Home Screen and installable PWA.
- Offline-supported product, billing, inventory and AI workflows.
- Future Android/iOS packaging with Capacitor.
- Future Windows/macOS/Linux packaging with Tauri or Electron.

2.20 Security and Audit

- Secure authentication and role-based authorization.
- Session timeout and protected sensitive actions.
- Cost-price visibility and discount limits.
- Audit user, timestamp, action, old value, new value and reason.
- Local data protection and encrypted backups.

3. Complete Architecture and Overall Flow

The project must use a modular, scalable and production-oriented architecture. The user interface, business logic, data access, barcode services, reporting and AI components must be separated so that future retail categories, a centralized database, multi-store synchronization and mobile/desktop packaging can be added without rebuilding the application.

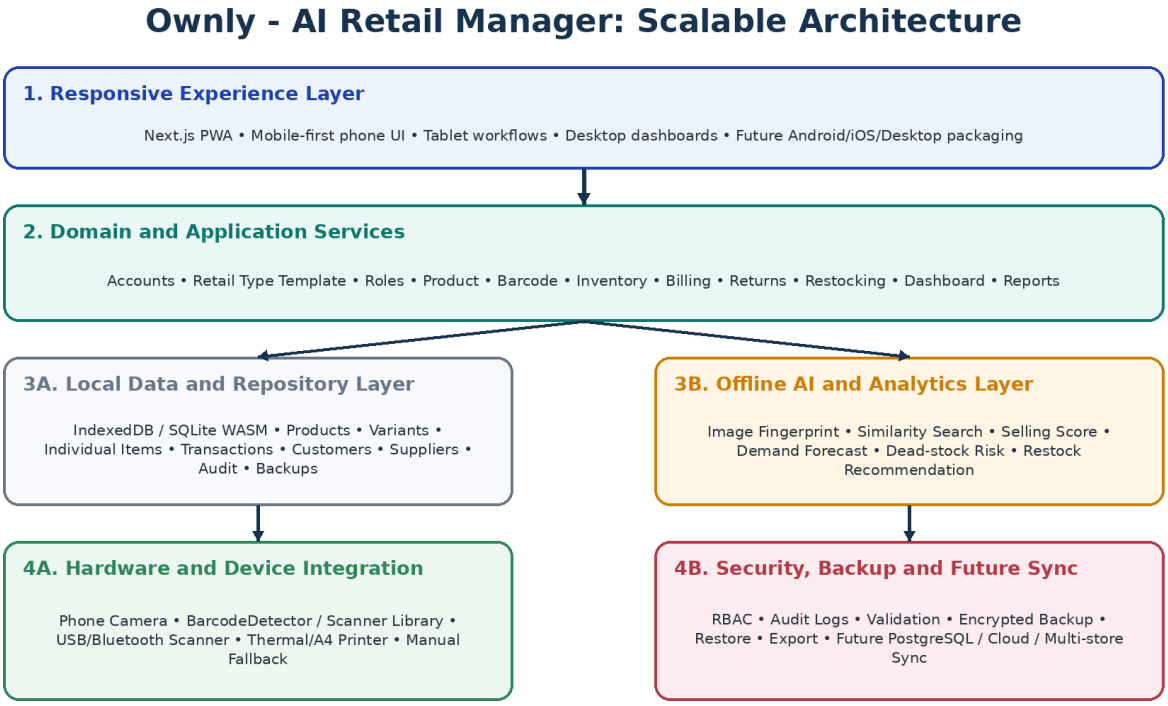


Figure 1. Scalable solution architecture.

3.1 Required Architecture Principles

- Feature/domain modules instead of one large application folder.
- Separate UI components, domain services, repositories and storage adapters.
- Centralized validation, error handling, logging and configuration.
- Configurable retail templates rather than clothing fields hard-coded everywhere.
- Independent barcode, reporting, backup and AI services.
- Database abstraction that can later move from IndexedDB/SQLite WASM to PostgreSQL.
- Versioned schemas, migrations and model/config versions.
- Reusable components, clear contracts and strong TypeScript typing.
- Automated tests for product, barcode, inventory, billing, return and restocking workflows.
- Private source repository, meaningful commits, reviews and release tags.

No Vibe-Coding Requirement

The project must not be a collection of unverified AI-generated code, copied snippets, temporary fixes or oversized components. Every contribution must be understood, reviewed, tested and integrated into a documented foundation. Navkar Solutions may reject a working demo if the code lacks maintainability, scalability, testing or documentation.

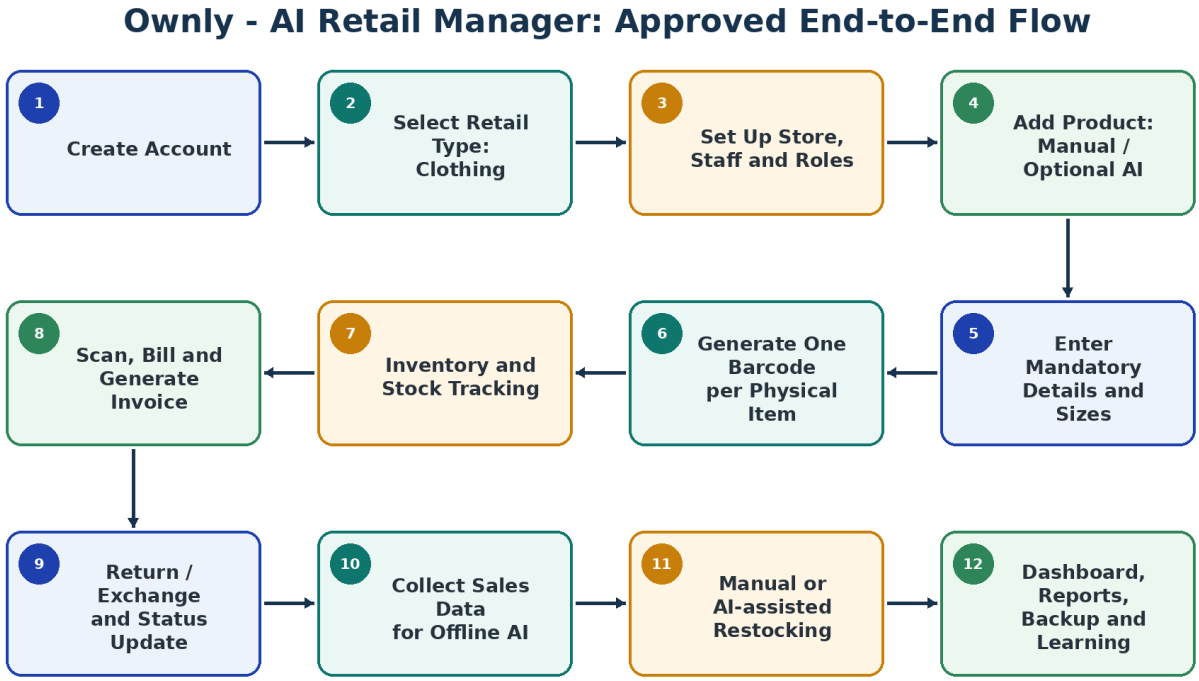


Figure 2. Approved end-to-end product and business flow.

3.2 Current Scope Boundary

Not Included in the Current Release

External social-media and marketplace research, including automated Instagram, Amazon, Meesho, Myntra or similar platform analysis, is excluded from the September delivery. The initial AI will learn from the retailer's own product, stock, sale, return, discount and restocking data.

4. Week-wise Development Plan

The student group must complete the entire approved project by 30 September 2026. The plan intentionally completes the non-AI MVP by the end of August, delivers AI functions by 20 September, and reserves the final ten days for integration, security, testing, documentation and handover.

Delivery Timeline: 30 July to 30 September 2026

Kickoff	30 Jul-2 Aug	Requirements, roles, UX and architecture
Week 1	3-9 Aug	Foundation, PWA, authentication and retail template
Week 2	10-16 Aug	Product, supplier, size-wise quantity and barcode
Week 3	17-23 Aug	Inventory, item status, search and stock movements
Week 4	24-30 Aug	Billing, invoices, returns and non-AI MVP
Week 5	31 Aug-6 Sep	Restocking, dashboards, reports and backup
Week 6	7-13 Sep	Optional photo AI, fingerprint and similarity
Week 7	14-20 Sep	AI scoring, demand, risk and restock advice
Week 8	21-27 Sep	Integration, security, testing and feature freeze
Handover	28-30 Sep	Final fixes, documentation, deployment and demo

Figure 3. Development timeline and hard deadlines.

Kickoff Period: 30 July-2 August 2026

Weekly objective: Confirm requirements, team ownership, design direction and scalable foundation.

Tasks

- Read and baseline the approved feature catalogue.
- Create private Git repository, branch policy, issue tracker and weekly-report template.
- Finalize technology stack and coding standards.
- Prepare mobile, tablet and desktop wireframes.
- Design architecture, folder structure, local data model and ER diagram.
- Define retail-template configuration with Clothing enabled.

Required deliverables

- Requirements baseline and scope exclusions.
- Team responsibility matrix.
- Responsive wireframes.
- Architecture and database diagrams.
- Repository and project board.
- Week 0 progress report.

Deadline / Review Gate

Approval Gate 1: Navkar Solutions approves requirements, architecture, data model and wireframes before feature development.

Week 1: 3-9 August 2026

Weekly objective: Build the scalable application foundation and onboarding.

Tasks

- Set up Next.js, TypeScript, PWA manifest and responsive application shell.
- Create domain-based folder structure, configuration and environment handling.
- Implement local database initialization and repository interfaces.
- Implement owner registration, login, store setup and retail-type selection.
- Enable Clothing and Fashion; show other retail types as Coming Soon.
- Implement staff accounts, roles and permission guards.
- Create navigation, layouts, validation, error handling and audit foundations.
- Write foundation unit tests and developer setup documentation.

Required deliverables

- Installable responsive application shell.
- Owner account and store onboarding.
- Retail-type selection.
- Staff and role management.
- Local data layer foundation.
- Architecture-compliant repository structure.

Deadline / Review Gate

Deadline: 9 August. Foundation review must confirm there is no hard-coded, tightly coupled or temporary project structure.

Week 2: 10-16 August 2026

Weekly objective: Complete product, supplier, size-wise stock and unique barcode creation.

Tasks

- Implement categories, subcategories and suppliers.
- Build Add Product with Manual and Optional AI Photo options.
- Make category, subcategory, supplier, cost, selling price and size-wise quantity mandatory.
- Create size checkboxes with quantity fields and automatic total calculation.
- Create Product, Product Variant and Inventory Item entities.
- Generate a variant SKU and one unique barcode per physical item.
- Build barcode-label preview, selective print, print-all and safe reprint.
- Add input validation and barcode uniqueness tests.

Required deliverables

- Product and supplier module.
- Size-wise quantity interface.
- Individual-item barcode engine.
- Thermal/A4 label templates.
- Test evidence: quantity N generates exactly N unique barcodes.

Deadline / Review Gate

Approval Gate 2: A 20-item product entry must generate exactly 20 valid and non-duplicated item barcodes.

Week 3: 17-23 August 2026

Weekly objective: Complete inventory and individual-item lifecycle tracking.

Tasks

- Build inventory list, detail view, search and filters.
- Implement Available, Sold, Returned, Exchanged, Damaged, Defective, Missing and Supplier Return statuses.
- Implement stock movements and immutable transaction history.
- Add category-, supplier-, size- and status-wise views.
- Calculate cost value, expected selling value and potential gross profit.

- Implement low-stock, out-of-stock and ageing rules.
- Prevent duplicate stock creation during barcode reprint.
- Add inventory tests, audit events and error cases.

Required deliverables

- Complete inventory module.
- Item-level status and movement history.
- Stock valuation and ageing.
- Search, filters and alerts.
- Inventory test report.

Deadline / Review Gate

Deadline: 23 August. Every item must be traceable from creation to its current status.

Week 4: 24-30 August 2026

Weekly objective: Deliver the complete non-AI MVP: billing, invoices, returns and exchanges.

Tasks

- Build barcode billing screen.
- Support phone camera, USB/Bluetooth scanner and manual barcode input.
- Automatically load product, size and selling price.
- Implement discount permissions, optional customer entry and payment modes.
- Generate thermal and A4 invoices.
- Mark scanned items Sold and update inventory transactionally.
- Implement return, exchange, refund/credit note, condition and invoice linking.
- Test invalid, duplicate, already-sold and returned barcode scenarios.

Required deliverables

- Working barcode billing.
- Cash, UPI, card and mixed payments.
- Invoice printing.
- Return and exchange module.
- Immediate stock updates and audit trail.

Deadline / Review Gate

Major Milestone: Complete non-AI MVP must be demonstrated and approved by 30 August 2026.

Week 5: 31 August-6 September 2026

Weekly objective: Add restocking, operational dashboards, reports and data protection.

Tasks

- Implement manual restocking with size-wise quantities.
- Generate new unique barcodes for restocked items and preserve history.
- Build dashboard version 1 with sales, profit, stock and quick actions.
- Add top sellers, slow movers, low stock, ageing and product performance.
- Implement sales, inventory, supplier, returns and staff reports.
- Implement CSV/Excel export.
- Implement full backup, restore, backup reminders and storage-health status.
- Test restore correctness using sample data.

Required deliverables

- Manual restocking.
- Dashboard version 1.
- Core reports and exports.

- Backup and restore.
- Storage-health and backup evidence.

Deadline / Review Gate

Approval Gate 3: The complete manual business workflow must work without AI or a paid database.

Week 6: 7-13 September 2026

Weekly objective: Implement optional product-photo intelligence and local similarity analysis.

Tasks

- Integrate a lightweight offline image encoder.
- Capture/upload photo and create visual fingerprint.
- Add photo-quality checks and optional attribute suggestions.
- Store fingerprint with product record.
- Implement cosine or equivalent similarity search.
- Detect possible duplicates and show similar historical products.
- Calculate time to first sale, sales velocity, sell-through, stock age and size demand.
- Document model source, version, limitations and offline runtime.

Required deliverables

- Optional product-photo capture.
- Offline visual fingerprint.
- Similar-product search.
- Duplicate warning.
- Product-performance calculation service.

Deadline / Review Gate

Deadline: 13 September. Manual product entry must remain unaffected when photo AI is disabled or unavailable.

Week 7: 14-20 September 2026

Weekly objective: Complete offline AI scoring and AI-assisted restocking.

Tasks

- Implement selling, demand, profitability, dead-stock and return-risk scores.
- Estimate first-sale probability, 30-day sales and expected sell-through.
- Recommend total restock quantity and size-wise distribution.
- Create Strong Buy, Controlled Buy, Trial, Hold and Avoid decisions.
- Show confidence, data sufficiency and written reasons.
- Allow owner to accept, modify, ignore or reject.
- Store prediction, recommendation, owner decision and later actual result.
- Add AI evaluation metrics and prediction-versus-actual dashboard.

Required deliverables

- AI product analysis.
- AI-assisted restocking screen.
- Explainable scores and confidence.
- Recommended quantity and size mix.
- Prediction history and evaluation.

Deadline / Review Gate

Major Milestone: All planned AI functionality must be feature-complete by 20 September 2026.

Week 8: 21-27 September 2026

Weekly objective: Freeze features and complete integration, security, quality and acceptance testing.

Tasks

- Stop adding new features after 20 September.
- Integrate and regression-test all modules.
- Review scalability, folder structure, dependency boundaries and technical debt.
- Complete RBAC, session controls, audit logs and protected sensitive actions.
- Test backup/restore, barcode integrity, inventory transactions and failure recovery.
- Test phone, tablet and desktop responsiveness.
- Test PWA installation and offline-supported workflows.
- Fix all critical and high-priority defects.
- Prepare user acceptance test evidence.

Required deliverables

- Integrated release candidate.
- Security and permission test report.
- Responsive and offline test evidence.
- Defect register and resolution status.
- Architecture and code-quality review.

Deadline / Review Gate

Hard Deadline: Feature-complete, tested release candidate must be ready by 27 September 2026.

Final Handover: 28-30 September 2026

Weekly objective: Submit the complete product, documentation and demonstration package.

Tasks

- Resolve final acceptance defects only.
- Deploy the final demonstration version.
- Tag the final source-code release.
- Submit setup, architecture, database, AI and user documentation.
- Submit test report, known limitations and future roadmap.
- Submit backup/restore file and demonstration data.
- Conduct final live demonstration and hand over credentials.
- Obtain formal acceptance from Navkar Solutions.

Required deliverables

- Final responsive PWA.
- Private source repository and release tag.
- All functional modules.
- Offline AI models/configuration.
- Technical documentation and user manual.
- Testing and acceptance evidence.
- Final presentation and live demo.

Deadline / Review Gate

Final Deadline: All deliverables must be submitted and accepted by 30 September 2026.

5. Milestones, Reporting and Project Governance

Deadline	Mandatory Milestone
2 August 2026	Requirements, architecture and wireframes approved.
9 August 2026	Scalable application foundation and onboarding complete.
16 August 2026	Product, size-wise quantity and unique barcode module complete.
23 August 2026	Inventory and item lifecycle complete.
30 August 2026	Complete non-AI MVP demonstrated.
6 September 2026	Manual restocking, dashboard, reports and backup complete.
13 September 2026	Optional photo AI and similarity analysis complete.
20 September 2026	All AI analysis and restocking features complete; feature freeze begins.
27 September 2026	Release candidate, testing and acceptance evidence complete.
30 September 2026	Final submission, demonstration and handover complete.

5.1 Weekly Progress Report

Arya Pawar, as Student Group Head, must submit one consolidated weekly progress report to Navkar Solutions every Saturday by 6:00 PM. The report must be factual and supported by repository links, screenshots, test results or a working demonstration.

- Work completed during the week.
- Individual contribution of every student.
- Modules and files changed.
- Architecture, schema or interface changes.
- Git commit and pull-request summary.
- Tests completed and results.
- Screenshots or demo evidence.
- Current issues, blockers and risks.
- Technical debt or temporary solutions introduced.
- Refactoring and scalability improvements completed.
- Plan and measurable targets for the following week.

5.2 Engineering Quality Conditions

- No major scope change after 2 August without written approval.
- No new feature after 20 September; only integration, testing and defect fixing.
- All AI-generated code must be reviewed, understood and tested.
- No hard-coded production data, passwords or business values.
- No direct data access from UI pages; use defined services/repositories.
- No duplicate business logic or unmaintainable oversized components.
- Critical workflows require unit/integration/end-to-end tests.
- The repository must remain private and accessible to Navkar Solutions.
- Architecture, setup, dependencies, environment variables and deployment must be documented.
- Unresolved defects and limitations must be disclosed honestly at handover.

6. Final Deliverables

No.	ID	Deliverable
1	D1	Approved requirements, scope and user stories.
2	D2	Responsive mobile, tablet and desktop wireframes/designs.
3	D3	Architecture, folder structure, ER/data model and security design.
4	D4	Working Next.js PWA and deployment.
5	D5	Account, retail-type selection, store setup, staff and roles.
6	D6	Product, category, subcategory and supplier modules.
7	D7	Size-wise quantity entry and one unique barcode per physical item.
8	D8	Inventory, status, stock movement, valuation and ageing.
9	D9	Billing, payment, invoice, return and exchange modules.
10	D10	Manual restocking with new barcodes.
11	D11	Offline product-photo fingerprint and similar-product analysis.
12	D12	Offline product-performance and AI-assisted restocking.
13	D13	Dashboard, filters, charts, reports and exports.
14	D14	Backup, restore, audit, security and storage health.
15	D15	Test plan, cases, results, defect log and acceptance evidence.
16	D16	Technical setup guide, user manual and administrator guide.
17	D17	Final presentation, live demonstration and handover package.

6.1 Final Acceptance Criteria

- A user can create an account, select Clothing and complete store onboarding.
- A product with multiple sizes can be entered using mandatory fields.
- The exact physical quantity produces the exact same number of unique barcodes.
- Every item is traceable through inventory, sale, return, exchange and restocking.
- Billing updates inventory correctly and produces an invoice.
- Manual restocking works without AI.
- Optional AI analysis provides explainable scores, confidence and recommendation.
- The application works responsively on phone, tablet and desktop.
- Backup can be exported and restored successfully.
- Role permissions, audit logging and error handling are demonstrated.
- Code structure, tests and documentation meet the scalable-engineering requirements.
- All deliverables are submitted by 30 September 2026.

Project Completion Certificate

Project Completion Certificates will be issued only after the student group submits all deliverables, completes the final demonstration, hands over the source code and documentation, fulfils weekly reporting requirements, resolves acceptance defects and receives formal approval from Navkar Solutions.